

APPLIED DATA SCIENCE CAPSTONE PROJECT

SpaceX Falcon 9 First-Stage Landing Prediction

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Machine learning · geospatial analytics · REST APIs · SQL · data visualization, github :[github]

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THE QUESTION

Landing success is predictable from mission context

Objective

Predict first-stage landing success to evaluate reusable-launch economics.

Pipeline

API and web data → wrangling → SQL and EDA → maps and dashboard → four classifiers.

Result

All four tuned models reached **83.33% test accuracy.**

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WHY RECOVERY MATTERS

Reusable boosters change the economics of launch

Falcon 9 recovery creates a reusable operating model where traditional launches discard expensive rocket stages.

Prediction problem

Estimate landing feasibility before launch from observable mission parameters.

Strategic value

Inform pricing, risk assessment, and launch-contracting decisions.

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DATA FOUNDATION

Two public data sources built the mission dataset

After filtering, the project isolated **90 operational Falcon 9 launches.**

SpaceX REST API

Paginated launch records enriched through rockets, payloads, launchpads, and cores endpoints.

Wikipedia tables

Flight number, date, booster version, site, payload, orbit, and customer metadata.

Connected ingestion pipeline

Two sources | Enrichment joins | Final launch table.

github :github



MODEL INPUTS

Feature engineering translated recovery outcomes into model inputs

01

Clean

Missing payload values were filled with central-tendency estimates.

02

Define

Class = 1: successful RTLS, ASDS, or ocean recovery.
Class = 0: failed or absent recovery.

03

Expand

One-hot encoding expanded orbit, site, landing pad, and booster serial into **83 predictors.**

github :github

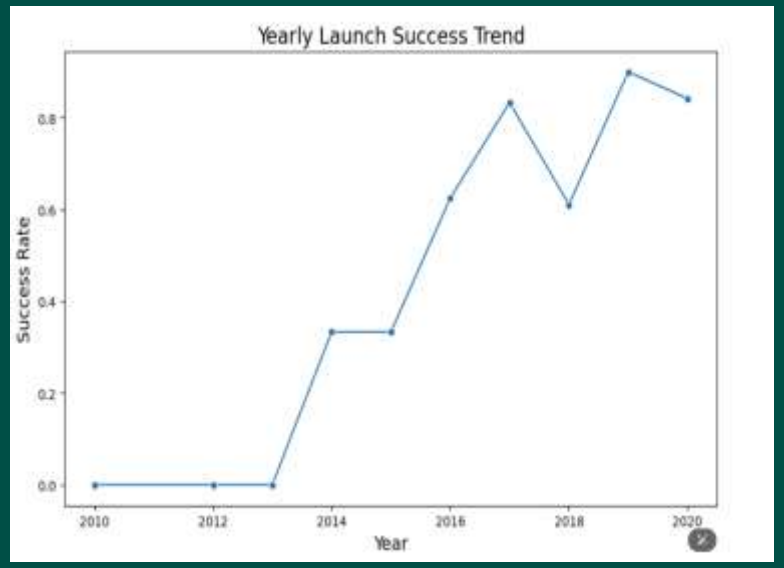
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EXPLORATORY ANALYSIS

Launch history reveals where recovery became repeatable

YEARLY TREND LINE

0% → >80%



SITE EXPANSION

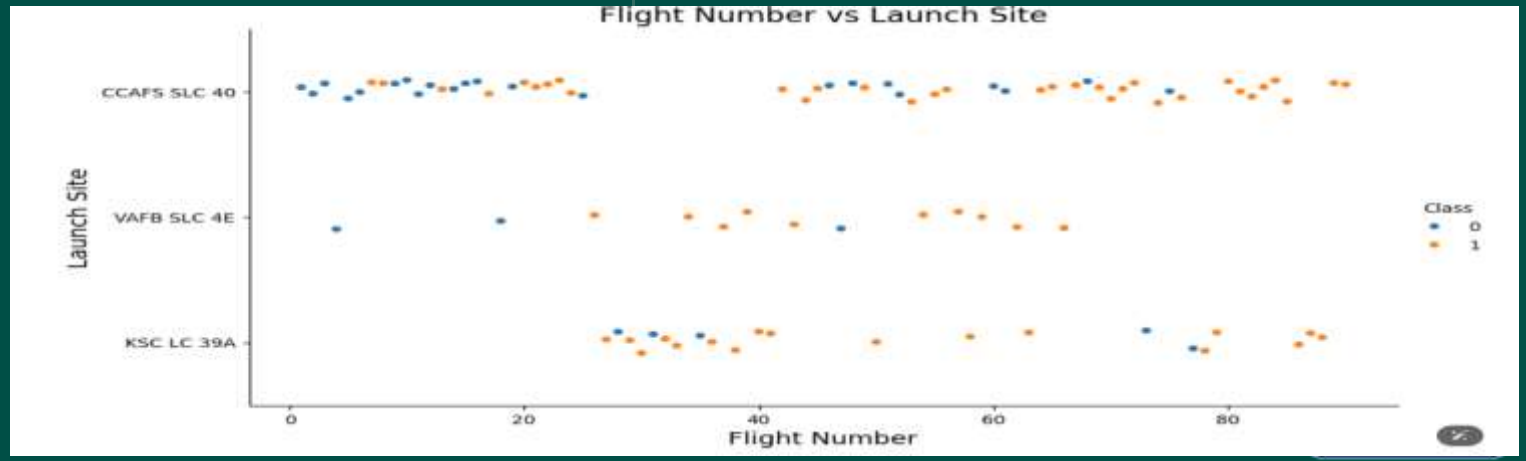
CCAFS SLC-40
VAFB SLC-4E
KSC LC-39A

Later missions show operational expansion across three launch sites.

PAYLOAD × SITE

KSC + CCAFS

Heavy payloads cluster at KSC and CCAFS; VAFB polar launches tend to carry lighter payloads.



github : gitub

QUERYABLE HISTORY

SQL made the mission history queryable

Representative findings from the launch database.

Query	Finding
Distinct launch sites	**4 sites**
NASA payload aggregation	**45,596 kg**
F9 v1.1 average payload	**2,928.4 kg**
Earliest ground landing	**2015-12-22**
Drone-ship payload range	**4,000–6,000 kg**

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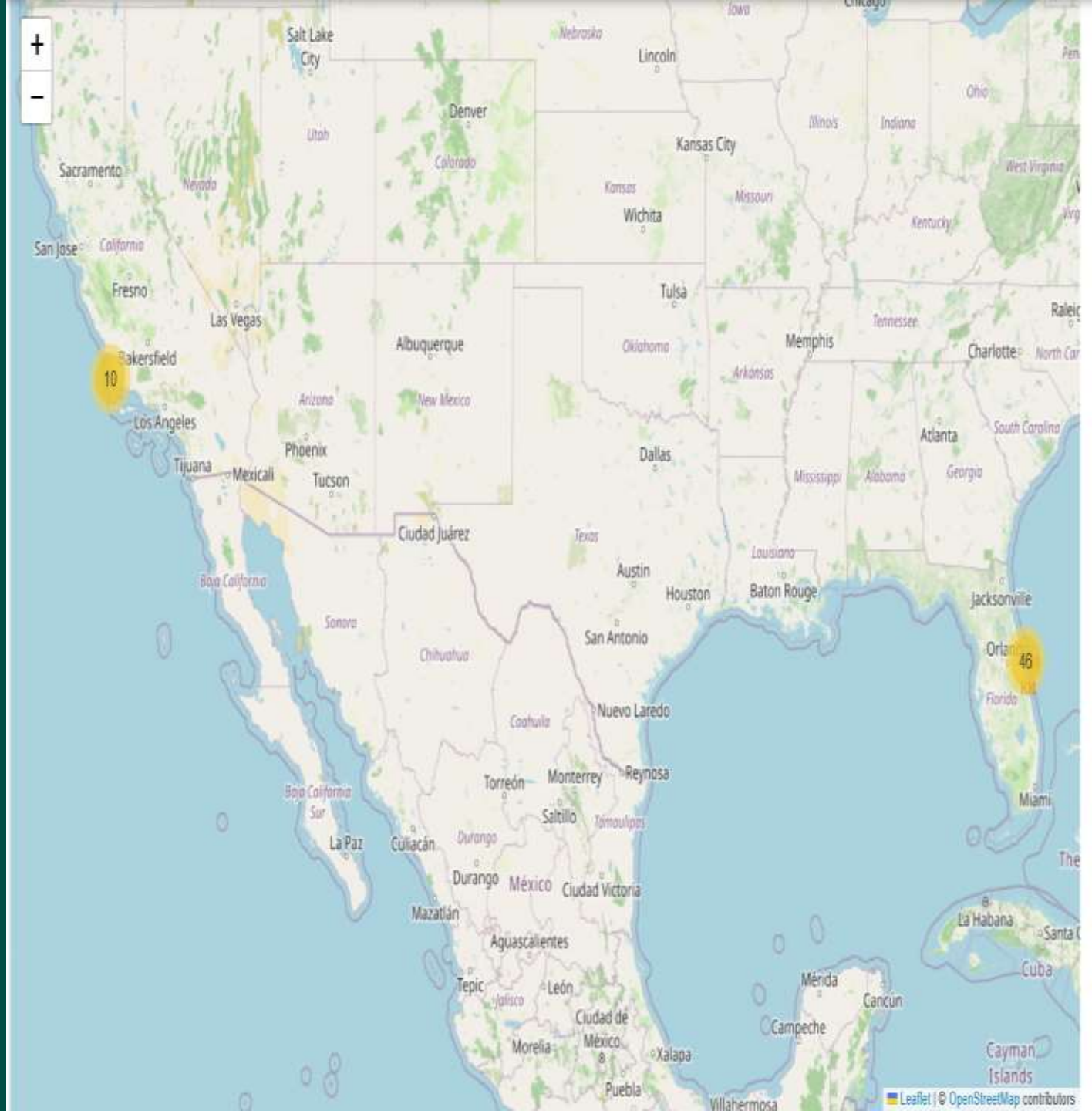
GEOSPATIAL ANALYTICS

Geospatial context explains recovery logistics

Folium mapped launch coordinates and colored outcomes by success or failure. Marker clusters kept dense launch histories legible across zoom levels.

SUCCESS FAILURE

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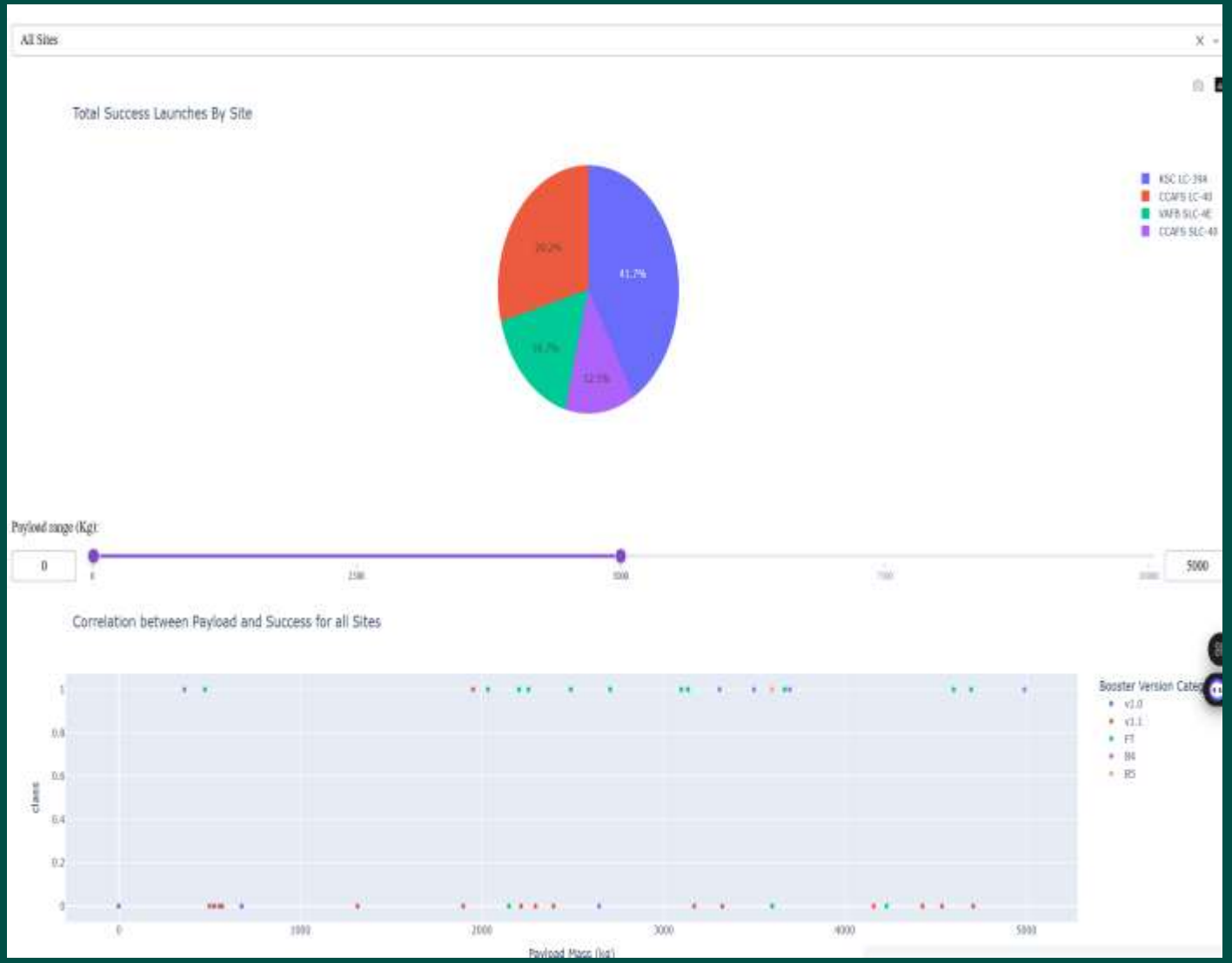
DECISION SURFACE

The dashboard turns the dataset into an interactive decision surface

Users filter by launch site and payload range from 0 to 10,000 kg.

Reactive outputs

Success versus failure proportions, plus payload mass, outcome, and booster version.



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MODEL COMPARISON

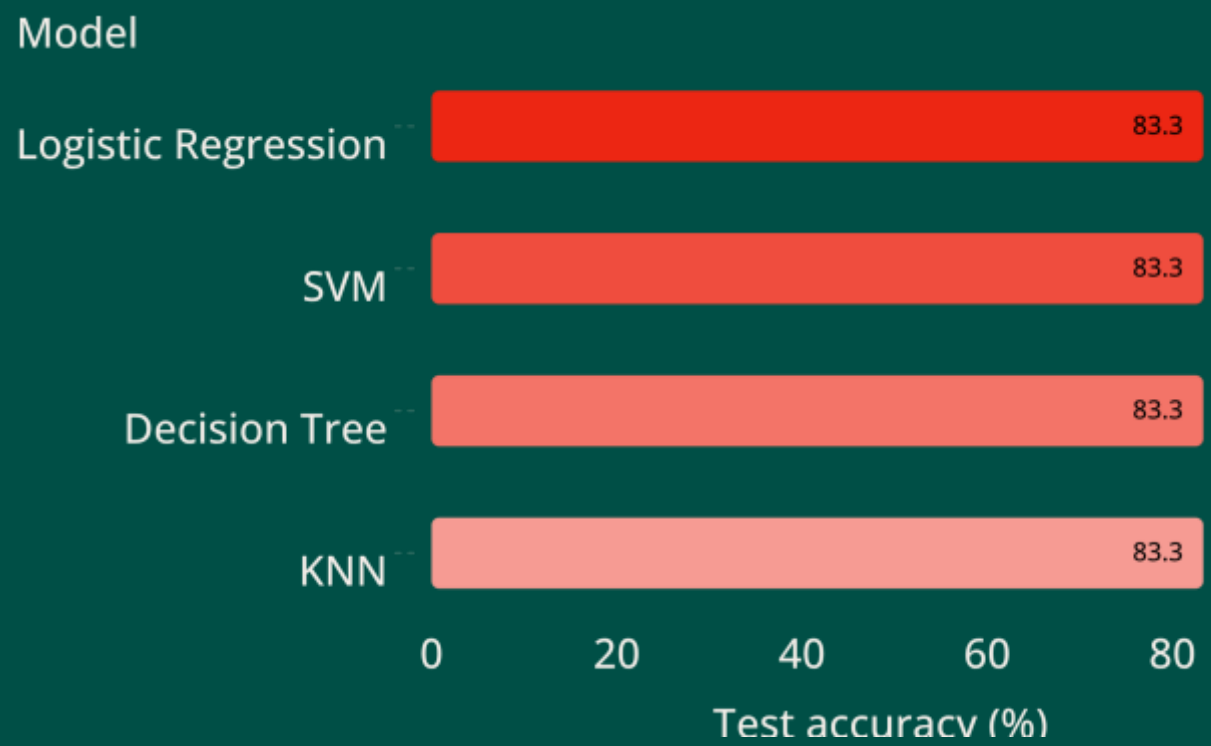
Four tuned classifiers converged on the same accuracy

Features were standardized and split 80/20 with `random_state=2`.

GridSearchCV used 10-fold cross-validation across Logistic Regression, SVM, Decision Tree, and KNN.

83.33%

Test accuracy for every tuned model.



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DATA FOUNDATION

Wikipedia tables complete the launch history

A structured web-ingestion path fills the historical gaps around the REST API dataset.

HTTP GET Request

HTML DOM Tree

BeautifulSoup Parsing

Row Extraction

Clean Pandas DataFrame

HTTP GET: requests fetch the raw Wikipedia HTML. **Parse:** BeautifulSoup navigates the nested DOM with `html.parser`. **Target:** select `wikitable plainrowheaders collapsible` tables.

Extract: rows and cells yield flight, booster, site, payload, orbit, customer, and outcome fields. **Clean:** remove citation markers and line breaks before assembling one Pandas DataFrame.

GitHub repository: [insert URL]

Outcome

Complete historical flight records complement and validate the REST API dataset.

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MODEL EVALUATION

Model evaluation supports a transparent deployment choice

CONFUSION MATRIX

Actual	Predicted success	Predicted failure
Success	TP 12	FN 0
Failure	FP 3	TN 3

Recall: 100% — no successful landing was misclassified as a failure.

Precision: ~80% — false positives are limited, with the small 18-record test split as context.

DEPLOYMENT VERDICT

Decision Tree

Accuracy: Logistic Regression, SVM, KNN, and Decision Tree each reached 83.33% on the test set.

Recommendation: deploy the Decision Tree for its explicit, rule-based paths and stakeholder-friendly interpretability.

Business impact: payload mass, destination orbit, and launch site can help estimate reuse probability and support more competitive launch bids.

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